9-1 Database [1]

Point!

1 Database Management System

(1) ([1]**Database**) : An organized collection of data, organized and stored in a format that makes it easy to access for specific purposes. In addition to collecting data, it also makes it easy to search, process, and share.
<Examples> Smartphone contact lists, company customer information, etc.

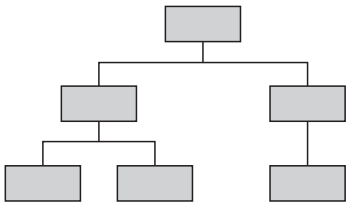
(2) ([2]**Database management system (DBMS)**) : A system that creates, operates, and manages databases.

(3) Database Management System Functions
[1] Data ([3]**consistency**) : Ensuring that concurrent operations for shared data do not cause inconsistencies.
[2] Data ([4]**integrity**) : Preventing duplication, tampering, and unauthorized registration or updating of data.
[3] Data ([5]**independence**) : Managing databases separately from the programs that use them.
[4] Data ([6]**confidentiality**) : Setting access permissions and performing authentication.
[5] Data ([7]**availability**) : Performing backups, restoration, and recovery to prepare for possible failures so that data can be recovered.

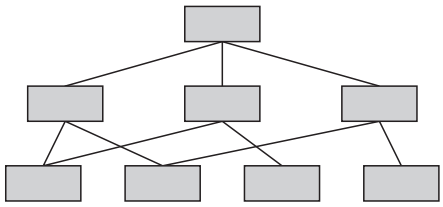
2 Types of Databases

(1) ([8]**Hierarchical database**) : Database where data is represented in a tree-like hierarchical structure.
(2) ([9]**Network database**) : Database where data is represented in a structure similar to a web or mesh.
(3) ([10]**Relational database**) : Database where collected data is organized and managed across multiple tables.

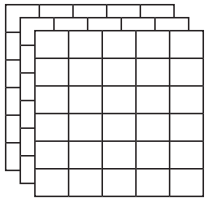
Hierarchical database



Network database



Relational database



(4) ([11]**NoSQL**) : Database management systems other than relational database management systems.

Warm Up

Answer the following questions.

(1) What is the term for information that is organized and stored in a format that makes it easy to access for a specific purpose?

(2) Choose one correct statement regarding data integrity from the options **A** to **D** below, and answer using the letter.

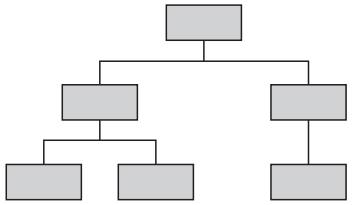
A Function where a database is managed separately from the programs that use it.

B Function that grants access rights to the data being used and limits users.

C Function that prevents duplication, tampering, and unauthorized registration or updating of data.

D Function for backing up and restoring to prepare for data failures.

(3) What is the term for a database with a structure like the one shown on the right?



Explanation

(1) Database

(2) **C**

(3) Data is represented in a tree-like hierarchical structure, therefore hierarchical database

Try

1 Answer the following questions.

(1) Complete the following sentences by filling in the blanks [1] to [4] with the appropriate terms.

> Data that is managed and stored in a certain format to make it easier to use large amounts of data is referred to as a ([1]). A system that creates, operates, and manages a ([1]) is called a ([2]), and it has functions such as data consistency, integrity, independence, ([3]), and ([4]).

(2) Choose the correct explanation for each of the terms [1] to [5] related to database management systems from the options **A** to **E**, and answer using the letters.

[1] Data independence [2] Data integrity [3] Data consistency
[4] Data availability [5] Data confidentiality

A Setting access permissions and performing authentication for data.
B Providing a mechanism to ensure there are no inconsistencies in the relationships between data.
C Method where stored data is managed separately according to its use.
D Backing up and taking other measures so that it is easier to recover data after a failure.
E Providing a mechanism to prevent accidental revisions to or unauthorized tampering of data.

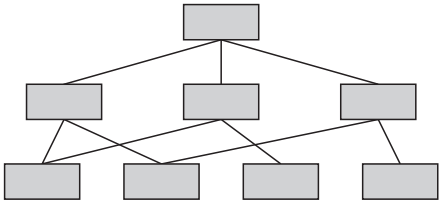
2 Answer the following questions.

(1) Choose the term that best fits into the blanks [1] to [3] in the following sentences from the options **A** to **D** below, and answer using the letters.

> There are several types of databases, including a ([1]) where collected data is organized and managed across multiple tables. Data that is organized in a hierarchical structure resembling a tree is referred to as a ([2]), while data expressed in a network-like structure is referred to as a ([3]).

A Hierarchical database **B** Relational database
C Network database **D** Object-oriented database

(2) What is the term for a database with a structure like the one shown on the right?



(3) What is the term for database management systems other than relational database management systems?

Exercise

1 Answer the following questions.

(1) Complete the following sentences by filling in the blanks [1] and [2] with the appropriate terms.

> A ([1]) refers to data that is organized and stored in a certain format for large-scale use. A ([1]) not only makes data easy to store, but also to search, process, and share. A ([2]) is a system for creating, operating, and managing a ([1]).

(2) Choose one correct statement regarding data availability from the options **A** to **D** below, and answer using the letter.

A Function where a database is managed separately from the programs that use it.
B Function that grants access rights to the data being used and limits users.
C Function that prevents duplication, tampering, and unauthorized registration or updating of data.
D Function for backing up and restoring to prepare for data failures.

(3) For the following statements [1] to [5] regarding the functions of database management systems, mark "○" if the statement is correct, and "×" if it is incorrect.

[1] Data confidentiality involves setting data access rights and implementing authentication.
[2] Data integrity means ensuring that concurrent operations for shared data do not cause inconsistencies.
[3] Data consistency involves not only preventing data duplication but also protecting against data tampering and the registration of fraudulent data.
[4] Data availability involves performing backups, restoration, and recovery to prepare for data failures.
[5] Data independence refers to managing databases separately from the programs that use them.

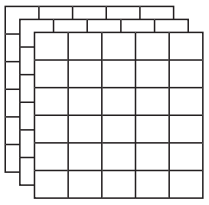
2 Answer the following questions.

(1) Choose the appropriate database for the following items [1] to [3] from the options **A** to **C** below, and answer using the letters.

[1] A database that represents data in a mesh-like structure.
[2] A database where collected data is organized and managed across multiple tables.
[3] A database where data is represented in a tree-like hierarchical structure.

A Hierarchical database **B** Network database
C Relational database

(2) What is the term for a database with a structure like the one shown on the right?



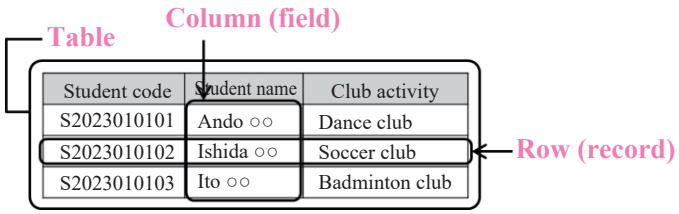
9-2 Database [2]

Point!

Relational Database

(1) ([1]**Relational database** (**RDB**)) : Database where collected data is organized and managed across multiple tables.

- A spreadsheet ([2]**table**) consists of ([3]**rows** (**records**)) and ([4]**columns** (**fields**)).



- Establishing ([5]**relationships**) between multiple tables makes it possible to eliminate duplicate data, which allows data to be handled with integrity.

(2) ([6]**SQL**) : A language used in relational databases to manipulate data.
Performs data registration, insertion, retrieval, and deletion.

(3) Relational Database Operations (Relational Algebra Operations)

[1] ([7]**Selection**) : Only rows that meet given conditions are extracted and displayed.

Student code	Student name	Club code
S2023010101	Ando ○○	C1
S2023010102	Ishida ○○	C2
S2023010103	Ito ○○	C1
S2023010104	Udagawa ○○	C1
S2023010105	Ezawa ○○	C2

Record for club code C2 → **Selection**

Student code	Student name	Club code
S2023010102	Ishida ○○	C2
S2023010105	Ezawa ○○	C2

[2] ([8]**Projection**) : Displaying only certain columns from a table.

Student code	Student name	Club code
S2023010101	Ando ○○	C1
S2023010102	Ishida ○○	C2
S2023010103	Ito ○○	C1
S2023010104	Udagawa ○○	C1
S2023010105	Ezawa ○○	C2

Student name → **Projection**

Student name
Ando ○○
Ishida ○○
Ito ○○
Udagawa ○○
Ezawa ○○

[3] ([9]**Join**) : Linking data from multiple tables according to specific conditions and displaying it as a single table.

Club table

Club code	Club activity
C1	Dance club
C2	Tennis club

Student table

Student code	Student name	Club code
S2023010101	Ando ○○	C1
S2023010102	Ishida ○○	C2
S2023010103	Ito ○○	C1
S2023010104	Udagawa ○○	C1
S2023010105	Ezawa ○○	C2

Join

Student code	Student name	Club code	Club activity
S2023010101	Ando ○○	C1	Dance club
S2023010102	Ishida ○○	C2	Tennis club
S2023010103	Ito ○○	C1	Dance club
S2023010104	Udagawa ○○	C1	Dance club
S2023010105	Ezawa ○○	C2	Tennis club

Warm Up

Answer the following questions.

(1) Complete the following sentences by filling in the blanks [1] to [4] with the appropriate terms.

> Summarizing data into a table often makes it easier to understand. A database where data is managed in a table is referred to as a ([1]). In a ([1]), rows are referred to as ([2]), and columns are referred to as ([3]). Also, establishing ([4]) between multiple tables makes it possible to eliminate duplicate data, which allows data to be handled with integrity.

(2) Consider the operations of relational databases. Table 3 below was created by performing certain operations with Table 1 and Table 2. List all the operations performed among selection, projection, and join.

Table 1

Student code	Student name	Club code
S2023010101	Ando ○○	C1
S2023010102	Ishida ○○	C2
S2023010103	Ito ○○	C1
S2023010104	Udagawa ○○	C1
S2023010105	Ezawa ○○	C2

Table 2

Club code	Club activity
C1	Dance club
C2	Tennis club

Table 3

Student code	Student name	Club activity
S2023010101	Ando ○○	Dance club
S2023010102	Ishida ○○	Tennis club
S2023010103	Ito ○○	Dance club
S2023010104	Udagawa ○○	Dance club
S2023010105	Ezawa ○○	Tennis club

Explanation

(1) [1] Relational database (RDB) [2] Record [3] Field [4] Relationships

(2) Table 3 combines “Student code” and “Student name” from Table 1, and “Club activity” from Table 2, indicating that a join operation has been performed. Also, since the “Club code” in Table 1 is not included in Table 3, it can be concluded that it also involves a projection.
Therefore, join and projection

Try

Answer the following questions.

(1) Choose the appropriate name of the database where data is managed as a table from the options **A** to **C** below, and answer using the letter.

A Hierarchical database **B** Network database

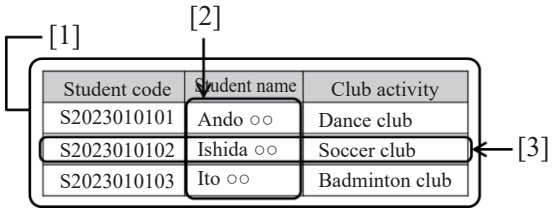
C Relational database

(2) Choose two appropriate names for each of the sections [1] to [3] in the table on the right from the options **A** to **F** below, and answer using the letters.

A Spreadsheet **B** Row

C Column **D** Field

E Table **F** Record



(3) Choose the term that best fits into the blanks [1] to [3] in the following sentences from the options **A** to **D** below, and answer using the letters.

Searches in a relational database are conducted using relational algebra operations.

- The process of extracting multiple columns from a table to create a new table is referred to as ([1]).
- The process of extracting only the rows that meet given conditions to create a new table is referred to as ([2]).
- The process of linking multiple tables based on the relationship between certain items to create a new table is referred to as ([3]).

A Projection **B** Join **C** Selection **D** Normalization

(4) For tables [1] to [3] which were created using the following two tables (Student table and Club table), choose the name of the operation that was performed from the options **A** to **D** below, and answer using the letters.

Student table

Student code	Student name	Club code
S2023010101	Ando ○○	C1
S2023010102	Ishida ○○	C2
S2023010103	Ito ○○	C1
S2023010104	Udagawa ○○	C1
S2023010105	Ezawa ○○	C2

Club table

Club code	Club activity
C1	Dance club
C2	Tennis club

[1]

Student name
Ando ○○
Ishida ○○
Ito ○○
Udagawa ○○
Ezawa ○○

[2]

Student code	Student name	Club code
S2023010102	Ishida ○○	C2
S2023010105	Ezawa ○○	C2

[3]

Student code	Student name	Club code	Club activity
S2023010101	Ando ○○	C1	Dance club
S2023010102	Ishida ○○	C2	Tennis club
S2023010103	Ito ○○	C1	Dance club
S2023010104	Udagawa ○○	C1	Dance club
S2023010105	Ezawa ○○	C2	Tennis club

A Projection **B** Join **C** Selection **D** Normalization

Exercise

1 Answer the following questions.

(1) Choose one correct statement regarding relational databases from the options **A** to **D** below, and answer using the letter.

A The operation of searching for and extracting records that meet specific conditions from a table is called "projection."

B A table consists of records and fields, and allows data to be managed in a table format.

C To maintain data integrity, it is not possible to link data from multiple tables.

D By displaying only specific columns from a table, it is possible to eliminate data redundancy.

(2) Choose one correct statement regarding SQL from the options **A** to **D** below, and answer using the letter.

A Mechanism that ensures the integrity of data stored in a relational database.

B Eliminating duplicate data from a relational database.

C Programming language for manipulating relational databases.

D Operations performed on a relational database such as selection, projection, and join.

(3) Choose one correct statement regarding the "join" operation of a relational database from the options **A** to **D** below, and answer using the letter.

A Creating a new table by linking information from multiple tables based on certain conditions.

B Using a language called SQL to register or delete data in a table.

C Extracting only the records from a table that meet specific conditions.

D Extracting only the fields from information contained in a table that meet specific conditions.

(4) For tables [1] to [3] which were created using the following two tables (Employee table and Branch table), choose the name of the operation that was performed from the options **A** to **D** below, and answer using the letters.

Employee table

Employee number	Name	Department	Branch number
10001	Sato ○○	Sales department	101
10002	Suzuki ○○	General affairs department	101
10003	Sakai ○○	Accounting department	102
10004	Kato ○○	Human resources department	102

Branch table

Branch number	Branch name
101	Tokyo
102	Osaka
103	Kyoto
104	Sapporo

[1]

Employee number	Name	Department	Branch number	Branch name
10001	Sato ○○	Sales department	101	Tokyo
10002	Suzuki ○○	General affairs department	101	Tokyo
10003	Sakai ○○	Accounting department	102	Osaka
10004	Kato ○○	Human resources department	102	Osaka

[2]

Employee number	Name	Department	Branch number
10003	Sakai ○○	Accounting department	102
10004	Kato ○○	Human resources department	102

[3]

Name	Department
Sato ○○	Sales department
Suzuki ○○	General affairs department
Sakai ○○	Accounting department
Kato ○○	Human resources department

A Projection **B** Join **C** Selection **D** Normalization

9-3 Various Information Systems

Point!

1 Information Systems

(1) ([1]**TPS- Transaction Processing System**) : A system that records and processes the company's routine daily transactions

<Example> Point of Sales (POS) System, Airline Reservation System

(2) ([2]**MIS - Management Information System**) : A system that provides periodic reports to manageres to assist them in decision-making.

<Example> Monthly Sales Reports, Inventory Reports.

(3) ([3]**DSS - Decision Support System**): A system that supports manageres in making non-routine decisions through modeling and simulaions.

<Example> "What if" Analysis, Business scenario planning

(4) ([4]**EIS - Executive Information System**): A system that provides summarised information to executive level about the overall performance of the organization.

<Example> Executive Dashboards, KPIs

2 System Integration

(1) ([5]**Vertical Integration**): Linking systems at different levels of the organization (such as TPS with MIS).

(2) ([6]**Horizontal Integration**): Linking the systems at the same level across different departments (such as the sales system with the inventory system).

3 Enterprise Resource Planning (ERP)

(1) ([7]**Enterprise Resource Planning System**) : An integrated system that combines various business functions such as accounting, human resources, supply chain, and customer relationship mnagaement into a single system

Warm Up

Answer the following questions.

(1) Choose the term that best fits into the blanks [1] to [3] in the following sentences from the options **A** to **E** below, and answer using the letters.

> Our daily lives are supported by ([1]) composed of humans, computers, and other information and communication technology equipment. Examples include ([2]) where data is collected and managed at the time a product is sold, and ([3]) where data from satellites is obtained to determine your present location and the operation status of buses and other transportation systems.

A POS system **B** Information systems **C** GPS (Global Positioning System)
D ITS (Intelligent Transport System) **E** SNS (Social Networking Service)

Explanation

(1) [1] **B** [2] **A** [3] **C**

Try

1 Answer the following questions.

(1) Complete the following sentences by filling in the blanks [1] to [4] with the appropriate terms.

> A system that utilizes a network to collect, share, and transmit various data and information is referred to as an (　[1]　). Examples of ([1]) include (　[2]　), which is a system that transmits and receives road information to resolve issues such as traffic accidents and congestion. In addition, a (　[3]　) is a system used in convenience stores and supermarkets to collect and manage data at the point of sales.

(2) What is the term for the satellite positioning system that collects location information for any place on Earth?

Exercise

1 Choose the most appropriate explanation of a POS system from the options **A** to **D** below, and answer using the letter.

A A system that uses artificial satellites to determine current locations.
B A system that operates by utilizing networks to collect, share, and transmit various data and information.
C A system used in convenience stores and other shops to manage product sales information.
D A system for transmitting and receiving road information to resolve issues such as traffic accidents and congestion.

2 Choose the most appropriate explanation for vertical integration from the options **A** to **D** below, and answer using the letter.

A Linking systems at the same level across different departments of the organization..
B Linking systems at different levels of the organization.
C Linking systems at a single level within a specific department.
D Linking systems between two unrelated departments